

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Computer Science and Engineering

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Lab Assignment

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Performing Convolution on a Music/Audio

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Objective

To apply a low-pass filter using convolution in order to smooth an audio signal and reduce high-frequency noise, demonstrating the fundamental concept of signal filtering in digital audio processing.

Theory

Audio signals are all around us—in music, conversations, podcasts, and digital media. However, these signals often come with unwanted noise or sharp, high-pitched elements that can be distracting. To clean and improve the sound, we use a technique called filtering. One popular method is the **low-pass filter**, which allows slower, smoother fluctuations (low frequencies) to pass through while reducing the impact of faster, more abrupt changes (high frequencies). The tool we use to apply this filter is **convolution**—a mathematical process where the audio signal is combined with a filter or “kernel” to produce a new, smoothed version. For example, a simple filter can be created by taking the average of nearby samples, effectively reducing sudden changes in amplitude. This process is foundational in digital signal processing and is used widely in applications ranging from music production to telecommunication. Moreover, software-based filtering makes high-quality audio processing more accessible and cost-effective, reducing the need for expensive physical hardware.

Mathematical Foundation

Convolution is a way to apply filters to signals. For continuous signals $x(t)$ and $h(t)$, it's defined as:

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

For discrete signals, it becomes:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n - k]$$

A simple low-pass filter like the moving average can be written as:

$$h[n] = \frac{1}{N}, \quad 0 \leq n < N$$

This smooths the signal by averaging nearby values, effectively reducing high-frequency noise.

Implementation

We used Python in Google Colab to implement the filter. The following libraries helped process, filter, visualize, and play audio:

- `librosa` – for loading and displaying audio
- `scipy.signal` – for applying convolution
- `matplotlib` – for plotting waveforms

To keep things manageable, the audio clip used was limited to 10 seconds.

Code Example

```
import librosa
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import convolve
import soundfile as sf
# Loading audio - "brahms"
audio_path = librosa.ex('brahms') # classical music
y, sr = librosa.load(audio_path, duration=40.0)
# Designing a low-pass filter
kernel_size = 15
low_pass_kernel = np.ones(kernel_size) / kernel_size
# Applying convolution
y_filtered = convolve(y, low_pass_kernel, mode='same')
# Plotting original vs. filtered signal
plt.figure(figsize=(14, 6))
plt.subplot(2, 1, 1)
plt.plot(y, color='blue')
plt.title("Original Audio Signal - Brahms")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.subplot(2, 1, 2)
plt.plot(y_filtered, color='green')
plt.title("Filtered Audio Signal (Low-pass) - Brahms")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
```

```
plt.tight_layout()
plt.show()
```

Listing 1: Applying a Low-Pass Filter to an Audio Signal

Results and Observations

- The original signal contained a wide range of frequencies.
- After filtering, the waveform looked smoother and more consistent.
- Harsh high-frequency sounds were noticeably reduced.
- The filtered audio sounded softer and more polished, with less hiss and sharpness.

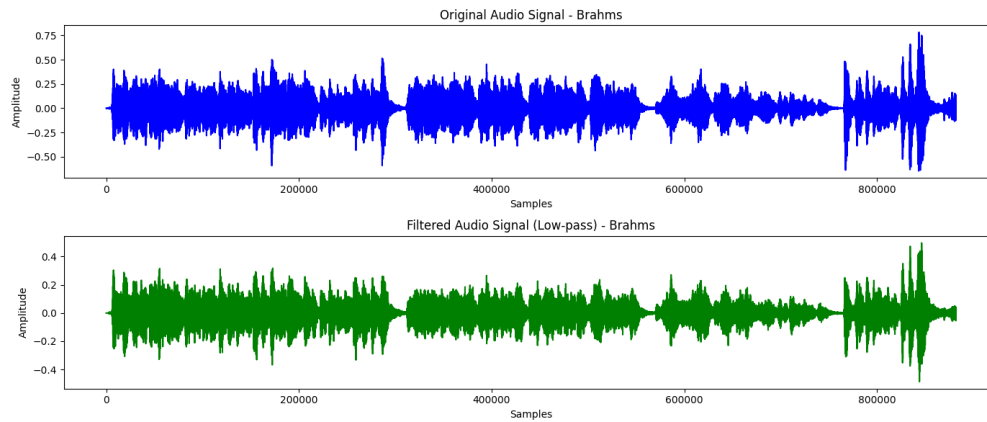


Figure 1: Convolution in Action

Conclusion

This project demonstrated how convolution can be used to apply a simple low-pass filter to an audio signal. The resulting smoothed waveform highlights the power of this basic yet essential technique. Such filters form the foundation for more advanced audio processing tools used in media production, broadcasting, and digital communication.